

Jaime Garrido *Engine & Graphics Programmer (C++)*

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SUMMARY

C++ engine and graphics programmer who has been building game engines and real-time rendering systems since 2021, with professional C/C++ experience since 2025 in safety-critical aerospace software. Currently co-developing a networked open-world multiplayer engine in C++ and OpenGL 4.5, with a deferred PBR renderer, DDGI global illumination and a Dear ImGui editor. Also built two custom engines, one of them on PlayStation 5 devkits, and gameplay systems in Unreal Engine 5 C++ for a game released on Steam. Looking for Graphics, Engine or Gameplay Programmer roles.

ENGINE & GRAPHICS PROJECTS

Open-World Multiplayer Game Engine

06/2026 - Present

Engine & Graphics Programmer | Independent project, co-developed with a second engineer | C++, OpenGL 4.5

- Built a networked client-server engine from scratch; one CMake codebase produces the dedicated server, game client, editor and a Tracy profiling build.
- Built a deferred PBR renderer (Cook-Torrance) with DDGI probe-based global illumination, hardware PCF shadows and reflective shadow maps, using compute shaders for GI atlas processing.
- Implemented DDGI probe settling: probes stop updating once lighting converges and only refresh when it changes, **reducing the GI update cost by 93%**, from 0.78 ms to 0.05 ms per frame (4,500 probes, 1080p).
- **Cut GI probe encoding from 665 μ s to 14 μ s per probe** by replacing a brute-force convolution with mipmap-based sampling.
- Implemented automatic LODs with meshoptimizer based on screen-space error, with separate LOD policies for the shadow and GI capture passes, plus per-submesh frustum culling.
- Built the editor with Dear ImGui (docking, multi-viewport) on top of an EnTT Entity-Component-System.

GUSSI Engine

09/2023 - 05/2025

Graphics Programmer | Custom engine, started as an academic project and continued solo | C++, OpenGL

- Built deferred and forward rendering pipelines with a G-buffer, SSAO, shadow-mapped dynamic lights and a post-processing stack.
- Wrote a multithreaded job system for asynchronous asset streaming, plus an ECS, a logger and a real-time ImGui editor.
- Implemented [Distance-Based Interior Mapping](#), replacing 350k+ vertices of room geometry with a parallax technique and **raising the frame rate from 335 to 4,400 FPS**.

Chee22, PS5 Devkit Game and Custom Engine (graded First Class)

02/2025 - 05/2025

Graphics & Gameplay Programmer | Academic project, Sheffield Hallam University | 5 programmers, 1 designer | C++, PS5 SDK

- Co-built a data-oriented, JSON-driven C++ engine for PlayStation 5 devkits by merging several of our student engines, following Sony's TRC requirements.
- Implemented a PBR rendering system and a post-processing pipeline driven by Python reflection that stacks effects such as posterization and pixelation in any order.
- Programmed the orbital turn-based camera and the core gameplay: piece movement, board logic and turn resolution.

Katto: Rising Tides, Arcadia Games

09/2023 - 07/2024

Gameplay Programmer | Academic project, 17-person team | Unreal Engine 5.2, C++ | Released on Steam, GDWC 2024 finalist

- Implemented core gameplay systems, world interactions and modular game logic in C++.
- Built spline-based real-time camera systems and shader-driven VFX, and profiled and optimized assets.

PROFESSIONAL EXPERIENCE

Software Verification Engineer, Embention

11/2025 - Present

Alicante, Spain | Safety-critical aerospace software (DO-178C)

- Verify and validate flight-critical firmware in a large embedded C/C++ codebase targeting the TI C2000 processor.
- Analyze C/C++ source and compiler-generated output for correctness, memory behavior and real-time performance on a resource-constrained target.
- Design and run unit, integration and structural coverage tests with VectorCAST under DO-178C and the JSF AV C++ coding standard, in a Git-based workflow with code review.

EDUCATION

BSc (Hons) Computer Science for Games, Sheffield Hallam University, Sheffield, UK

09/2024 - 07/2025

First Class Honours (highest UK degree classification). Final top-up year in advanced C++, software architecture and real-time systems.

Higher National Diploma (HND) in Computing, ESAT, Valencia, Spain

10/2021 - 07/2024

Distinction (highest grade). Three-year program in C/C++, OpenGL, low-level programming and engine architecture.

SKILLS

Programming: C++ (17/20/23), C, C#, Python, SQL, ARM Assembly

Graphics: OpenGL 4.5, GLSL, HLSL, PSSL, PBR, deferred and forward rendering, global illumination (DDGI), shadow mapping, SSAO, compute shaders, LOD, post-processing

Engine: engine architecture, ECS (EnTT), data-oriented design, multithreading and job systems, client-server networking, gameplay programming, editor tools (Dear ImGui)

Engines and platforms: Unreal Engine 5 (C++), Unity, PlayStation 5 SDK

Tools: Tracy, RenderDoc, NVIDIA Nsight, Razor CPU/GPU, Visual Studio, CMake, Git, Perforce, Jira, VectorCAST

Spoken languages: Spanish (native), English (C1)